

Eminence 6.0 – Network Simulation Task

Scenario

The university has a new campus with two buildings. You have been hired as the junior network engineer and asked to build the whole campus network in Cisco Packet Tracer using 3-tier hierarchical design.

Each department must live in its own VLAN so that broadcast traffic stays small. Inside each building the multilayer distribution switch routes between its own VLANs, and the core router carries traffic between the two buildings. Your job is to build the topology, separate the traffic at Layer 2, route between the VLANs at Layer 3, connect the two buildings through the core, and prove that everything works.

Equipment you need

Quantity	Device	Model to use in Packet Tracer	Role
1	Router	2911	Core router
2	Multilayer Switch	3560-24PS	Distribution switches
4	Switch	2960-24TT	Access switches
6	PC	PC-PT	End users

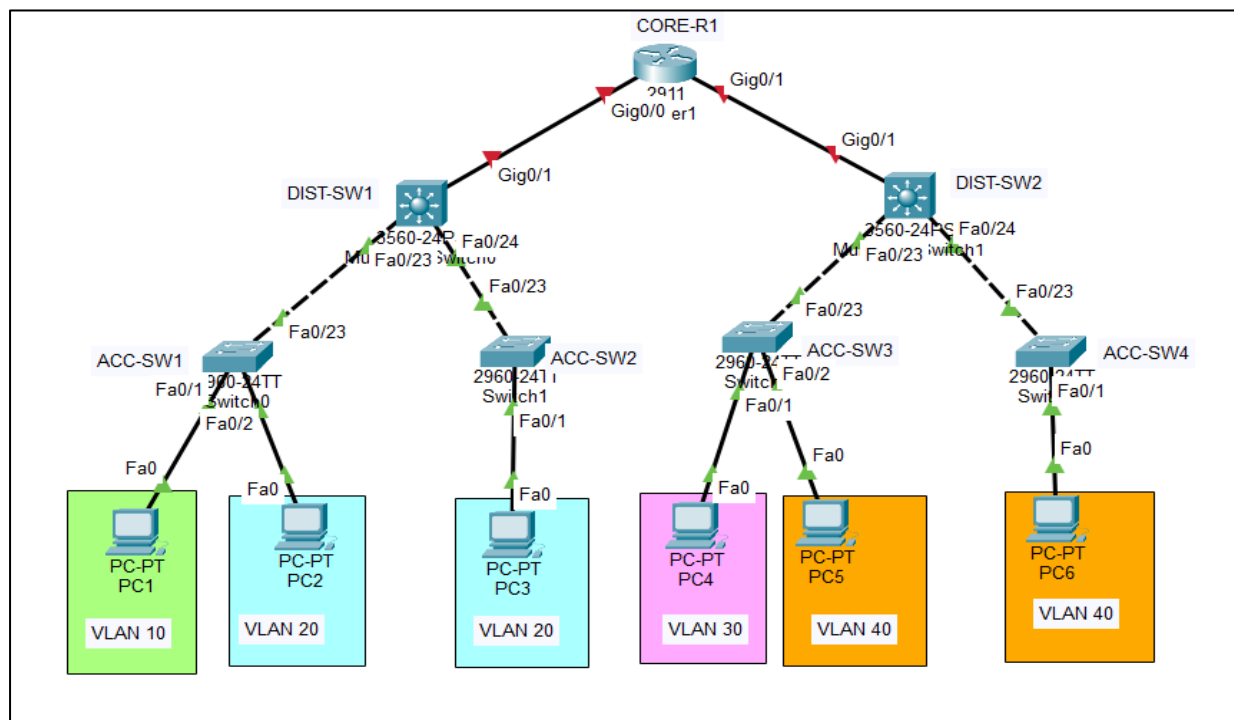


Figure 1: Campus Network Topology Diagram

Table 1: VLAN Plan

VLAN ID	VLAN Name	Purpose	Subnet	Gateway (SVI)	Lives in
10	FACULTY	Faculty staff	192.168.10.0/24	192.168.10.1	Building A
20	STUDENTS	Student labs	192.168.20.0/24	192.168.20.1	Building A
30	ADMIN	Administration	192.168.30.0/24	192.168.30.1	Building B
40	GUEST	Guest users	192.168.40.0/24	192.168.40.1	Building B

Table 2: Infrastructure Addressing

Device	Interface	IP Address / Mask	Notes
CORE-R1	Gi0/0	10.0.0.1 /30	Routed link to DIST-SW1
CORE-R1	Gi0/1	10.0.0.5 /30	Routed link to DIST-SW2
DIST-SW1	Gi0/1	10.0.0.2 /30	Routed port (no switchport)
DIST-SW1	SVI VLAN 10	192.168.10.1 /24	Default gateway for FACULTY
DIST-SW1	SVI VLAN 20	192.168.20.1 /24	Default gateway for STUDENTS
DIST-SW2	Gi0/1	10.0.0.6 /30	Routed port (no switchport)
DIST-SW2	SVI VLAN 30	192.168.30.1 /24	Default gateway for ADMIN
DIST-SW2	SVI VLAN 40	192.168.40.1 /24	Default gateway for GUEST

Table 3: End Devices Addressing

Device	VLAN	IP Address / Mask	Default Gateway
PC1	10	192.168.10.11/24	192.168.10.1
PC2	20	192.168.20.11/24	192.168.20.1
PC3	20	192.168.20.12/24	192.168.20.1
PC4	30	192.168.30.11/24	192.168.30.1
PC5	40	192.168.40.11/24	192.168.40.1
PC6	40	192.168.40.12/24	192.168.40.1

Table 4: Cabling Map

Device A	Port A	Device B	Port B
CORE-R1	Gi0/0	DIST-SW1	Gi0/1
CORE-R1	Gi0/1	DIST-SW2	Gi0/1
DIST-SW1	Fa0/23	ACC-SW1	Fa0/23
DIST-SW1	Fa0/24	ACC-SW2	Fa0/23
DIST-SW2	Fa0/23	ACC-SW3	Fa0/23
DIST-SW2	Fa0/24	ACC-SW4	Fa0/23
ACC-SW1	Fa0/1, Fa0/2	PC1, PC2	Fa0
ACC-SW2	Fa0/1	PC3	Fa0
ACC-SW3	Fa0/1, Fa0/2	PC4, PC5	Fa0
ACC-SW4	Fa0/1	PC6	Fa0

Part A – Basic Device Configurations

1. Drag in all devices listed above and cable them exactly as shown in Figure 1 and Table 4.
2. Using the CLI, set the hostname of every network device exactly as shown in Figure 1 and label them on the diagram.
3. Configure an enable secret password of “cisco123” on the router only.
4. Configure a login banner (MOTD) containing your team’s name on the two distribution switches only (DIST-SW1 and DIST-SW2).

Part B – VLANs and Access Ports (Layer 2)

5. Create the VLANs from Table 1 on every switch of both buildings.
6. Name each VLAN exactly as shown in Table 1.
7. Put every PC port into access mode and assign it to the correct VLAN.

Part C – Trunking (Layer 2)

8. Configure links between distribution switches to access switches as static 802.1Q trunks on both ends.
9. On the distribution switches you must set the encapsulation to dot1q before the port will accept trunk mode.
10. Restrict each trunk so that it only carries the VLANs that building actually uses (Building A: 10 and 20, Building B: 30 and 40).

Part D – Inter VLAN Routing on Multilayer Switches (Layer 3 Switching)

A PC in VLAN 10 cannot talk to a PC in VLAN 20 by switching alone - the traffic has to be routed. In this design each building's multilayer distribution switch does that job itself, using a Switch Virtual Interface (SVI) per VLAN.

11. Turn on routing on DIST-SW1 and DIST-SW2.
12. Create the SVIs listed in Table 2 on each distribution switch, give them their IP addresses and make sure they are up.
13. Configure PC1 to PC6 with the static addresses, masks and gateways from Table 3.
14. Verify that every PC can ping its own default gateway.
15. Verify inter-VLAN routing works inside each building. (PC1 (VLAN 10) can ping PC3 (VLAN 20) and that PC4 (VLAN 30) can ping PC6 (VLAN 40))

Part E – Core Layer Connectivity

16. Convert Gi0/1 on DIST-SW1 and Gi0/1 on DIST-SW2 into routed ports, then give them the /30 addresses from Table 2.

17. Configure CORE-R1 Gi0/0 and Gi0/1 with the addresses from Table 2 and bring both interfaces up.
18. Verify: CORE-R1 can ping 10.0.0.2 and 10.0.0.6.

Part F – Routing

Building A is connected to the core with a dynamic routing protocol. Building B is connected with static routing. The core router must join the two worlds together so that every PC can reach every other PC.

19. Configure OSPF between CORE-R1 and DIST-SW1.
20. Verify with show ip ospf neighbor - the neighbour state must be FULL.
21. Configure static routing between CORE-R1 and DIST-SW2.
22. Building A still cannot reach Building B, because static routes are not advertised automatically. On CORE-R1, redistribute the static routes into OSPF.

Part G – Verification and Analysis

Every test below must succeed in your saved file

#	Test	What it proves
1	Every PC can ping its own default gateway	VLAN, access port and SVI configuration
2	PC1 (VLAN 10) can ping PC3 (VLAN 20)	Trunking and inter-VLAN routing in Building A
3	PC4 (VLAN 30) can ping PC6 (VLAN 40)	Trunking and inter-VLAN routing in Building B
4	CORE-R1 can ping 10.0.0.2 and 10.0.0.6	Routed ports and core addressing
5	PC1 can ping PC4	OSPF, static routes and redistribution
6	PC2 can ping PC6	Full end-to-end connectivity across the campus
7	tracert from PC1 to PC4 shows 192.168.10.1, then 10.0.0.1, then 10.0.0.6	Traffic really follows the hierarchy

Submission

- Submit ONE Packet Tracer file (.pkt) to the Network Task folder in your team's GitHub repository.
- Rename the file as: **TeamName_NetworkTask.pkt**